

VTET: A Virtual Industrial Control System Testbed for Cyber Security Research

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Abstract—Nowadays Industrial Control System (ICS) are suffering the increasing cyber attacks. The ICS testbed is a ideal platform since that it could make the security research more controllable and safe. In contrast to the physically replication testbed, the virtual testbed is more economic, safe and maintainable which will makes it advantageous in the laboratorial environment. In this paper, we present *VTET*, a virtual ICS testbed for cyber security research. *VTET* contains a virtual chemical industrial process, and both virtual and physical controllers. We provide a detailed description of *VTET* implementation such as the migration of controllers and the configuration of network. And we also demonstrate the 5 attack schemes simulated in *VTET* to prove its availability. Since the convenience of configuration and deployment, *VTET* is very suitable for academic ICS security research.

Index Terms—Industrial Control System, Cyber Security, Virtual Testbed

I. INTRODUCTION

Industrial Control System (ICS) is the collection of several subsystems which providing the monitoring and controlling functions to the industrial processes. Depending on the different application scenarios, ICS can be implemented by Supervisory Control and Data Acquisition (SCADA) systems or Distributed Control Systems (DCS). These systems have a common architecture including the physical process, instrumentations, Human-Machine-Interface (HMI), controllers.

Conventional ICSs were considered to be immune to cyber attacks due to their proprietary networks and hardware [1]. Nowadays Information Technologies such as embedded devices and TCP/IP based protocols have been integrated into the modern ICSs. The vulnerabilities of the IT components would put ICSs at the risk of the cyber attacks. In the recent decade, the dramatic rise of ICS cyber incidents has promoted the researches on ICS cyber security.

Most cyber security researches, such as reproducing the cyber attack, analysing the vulnerability and evaluating the security enhancement, would result in unstability of the process and damages of the devices [2]. So the ICS testbed is proposed to perform the researches instead of the real-life ICS. The testbed is a duplicate or scaled-down ICS which could make the security experiments more controllable and safe.

To achieve the best fidelity, the testbed could be constructed by physically replicating the devices and process from a real-

world ICS. However there are still some disadvantages of the physical replication testbed: 1) It is expensive and large space-consuming, 2) Some processes are risky in laboratory (e.g., chemical reaction and high voltage transmission), 3) The experiments would be time-consuming and unrepeatable due to the real-life industrial process, and 4) It is hard to maintain and recover the system (including hardware and software) especially after suffering the cyber attacks.

Virtualization of the testbed is a straightforward approach to overcome the above disadvantages. Although the virtual testbed would lose some fidelity, it is suitable in laboratorial environment for preliminary ICS security research.

In this paper, we propose a virtual ICS testbed *VTET* (Virtual Tennessee-Eastman Testbed) for cyber security research. *VTET* contains the typical components (including process, controller and control network) in a chemical industrial control system. The process in *VTET* is the Tennessee-Eastman process (or TE problem) which is simulated by Matlab. The controller in *VTET* could be physical PLC (Programmable Logic Controller) or a simulator. *VTET* supports 3 common ICS protocols (OPC, Modbus and Siemens S7) for network communication. We have also implemented 5 attacks in *VTET* that could be used for ICS cyber security analysis.

The contributions of this paper are as follows:

- We propose a virtualization scheme for ICS testbed and implement a virtual testbed which could work on both full-virtualization and semi-virtualization modes (in Sec.II).
- We implement the migration of the controller from TE Simulink model to the PLC which would achieve more fidelity (in Sec.III-C).
- The network architecture of *VTET* could support 3 common industrial protocols which could enable to expand different devices (in Sec.III-D).
- We demonstrate 5 attacks on *VTET* to prove that *VTET* is available to experiment the attacks which are unfeasible to test in the physical replication testbed.

II. VTET OVERVIEW

As shown in Fig.1, *VTET* is comprised of 4 main components: a physical PLC, a PC used for network communication

and the rest two PCs simulating the process and PLC respectively. These components are connected by a router.

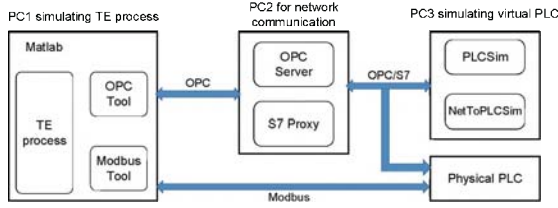


Fig. 1. The Architecture of VTET

VTET has 2 work modes: 1) In full-virtualization mode only PC3 is connected and the OPC and S7 protocols are supported; 2) In semi-virtualization mode, the physical PLC replaces PC3 and supports all 3 protocols.

VTET simulates a chemical process which is automate controlled by a PLC. The process runs in a cyclic manner. The general workflow in a single cycle is as follow:

- 1) PC1 generates the measurements of the process from the TE model and sends them to PC2;
- 2) PC3 or PLC queries the OPC sever or S7 proxy in PC2 for the measurements and generate the manipulated variables, then send them back to PC2¹;
- 3) The process in PC1 queries the manipulated variables from PC2, takes them as arguments to generates the measurements for the next cycle.

Since VTET simulating the physical parts of ICS, we can discover the PLC (simulated by PC3) through the network scan. This feature enable the researcher to perform the reconnaissance or PLC-targeted attack. In addition, VTET could supply multi-level of data for security analysis and assessment. The data of measurements and manipulated variables could be achieved in the Matlab model of PC1 and the raw data (in TCP/IP level) are collected by a sniffer in the router.

III. THE IMPLEMENTATION OF VTET

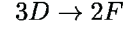
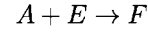
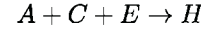
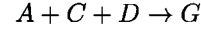
A. TE Process

As described above, the process in VTET is the well-known TE process which was proposed to test control technologies for continuous processes [3]. Since the TE process is an abstraction of a real-world chemical production process, it could reflect the impact of the cyber attack to an ICS. Moreover, TE process is nonlinear and continuous, so it is complex enough to evaluating diverse cyber attacks or protection methods.

We give a brief introduction of TE process firstly. The architecture of TE process is shown in Fig. 2. There are five components in TE: a two-phase reactor, a product condenser, a vapor-liquid separator, a product stripper, and a recycle compressor. The process produces two products, G and H , from four reactants A , C , D , and E . F is the byproduct.

¹In semi-virtualization mode, PC1 could directly connect to physical PLC using Modbus protocol, so in this way PC2 is no needed to transmit the data.

The equations of the process are presented as follows:



The gaseous reactants A , D , E are fed to the reactor to form liquid products. The reaction is exothermic, so the reactor temperature should be controlled using cooling bundles. The vaporized products of the reactor is then fed to a condenser to be cooled into liquid form again. The vapor-liquid separator is used to separate unreacted gases from the liquid products and sent them back to the reactor by a centrifugal recycle compressor. The remainder will mix with C in the stripper. The products, G and H , are stripped and sent downstream for further refining. Byproducts F are purged from the process.

There are 41 measurements and 12 manipulated variables in TE, which are used for applying the control strategy. We choose the control strategy proposed by Ricker [4]. This model is implemented as a Matlab Simulink model and is very easy to deploy. Fig. 3 is the TE model in Matlab.

B. Programmable Logic Controller

The virtualization of PLC is implemented using PLCSim² and NetToPLCSim³. PLCSim is the official simulator of Siemens PLC. PLCSim could simulate most functions of a real PLC, but it can't communicate with other components through the network. So we introduce NetToPLCSim to enable the network communication of PLCSim. With the help of NetToPLCSim, other componets (SCADA, HMI, etc.) would recognize the PLCSim as a real PLC and establish the connection with it.

C. Migrating TE Controller To PLC

The original TE Simulink model contains the controller itself. The TE controller should be separated and translated to the PLC program. Then the TE process will be stable under the control of PLC. This makes VTET more closer to an actual ICS.

In Matlab, when we explore inside the TE plant in Fig. 3, we could see the TE process and the controller as in Fig. 4.

As shown in Fig. 4(b), there are 20 control-loops and each one should be migrated as a function block to the PLC. We will take y_A control-loop as example to demonstrate how to migrate a control-loop to PLC. The y_A control-loop is shown in Fig. 5.

Matlab could directly transform the control-loop to PLC code. Firstly we should create a subsystem from y_A control-loop. Then we could transform this subsystem to a PLC code. The PLC used in VTET is Siemens S7-PLC, so we generate a Siemens format (the SCL format) code. The code

²<https://w3.siemens.com/mcms/simatic-controller-software/en/step7/simatic-s7-plcsim/Pages/Default.aspx>

³<https://sourceforge.net/projects/nettoplcsim/>

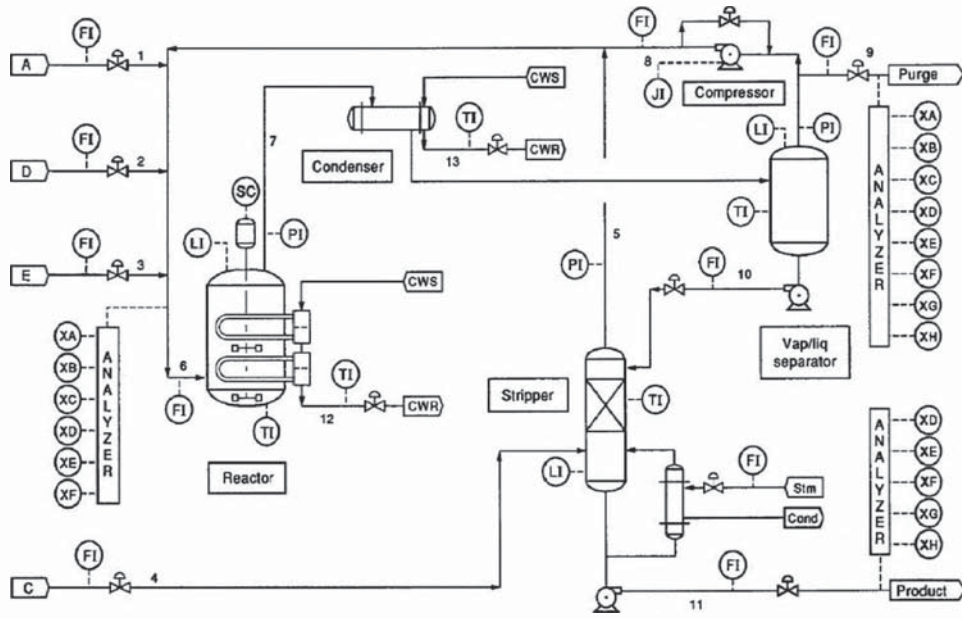


Fig. 2. Tennessee-Eastman process [3]

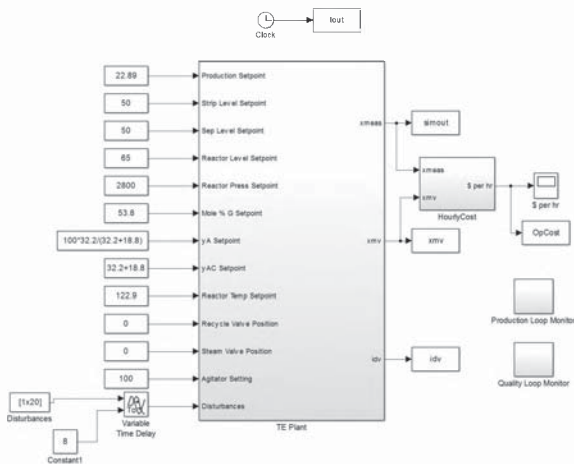


Fig. 3. The TE model

is then imported to TIA Portal (the program and configuration platform of Siemens) as shown in Fig. 6.

Then we should make some modifications of the code in TIA Portal. The code is transformed to a function block (e.g., FB25) and the 2 inputs and 1 output should also map to PLC variables.

Finally we invoke the function block in the OB35 (the organization block executed every 100ms) so that the yA control-loop could execute periodically.

After all the control-loops are migrated, we can download the code to the virtual and physical PLCs.

D. The Network

VTET supports 3 of the most widely-used ICS protocols: OPC, Modbus and Siemens-S7.

OPC (Open Platform Communications) is developed by the OPC Foundation which provides a standardized interface for the communication of HMI/SCADA and PLC [5].

Modbus is a standard communication protocol which is now a commonly available means of connecting industrial electronic devices.

S7 is a Siemens proprietary protocol that runs between PLC of the Siemens S7-300/400 family.

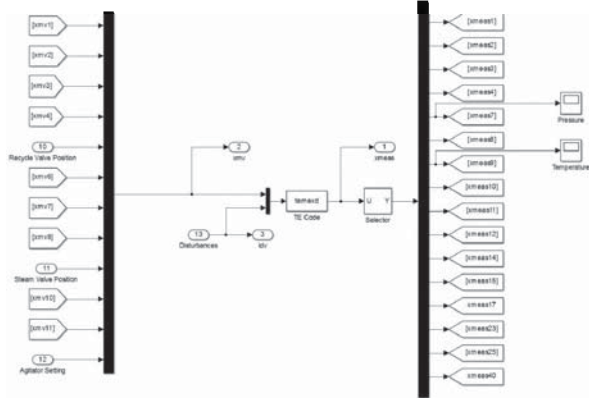
The configuration and implementation of network include two parts:

1) *Network Server (in PC2)*: In PC2, we install the OPC server (KEPServerEx) to play as the communication mediation of the process and PLC. The variables mapped to the process inputs and outputs should be configured in the server. Fig. 7 is the example of yA control-loop.

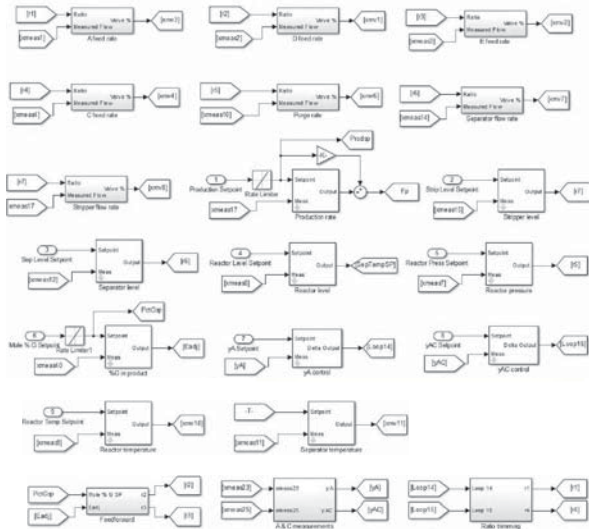
A S7 proxy is implemented using the Snap7 library⁴ (a library for S7 communication) for the communication through S7 protocol. The proxy periodically inquires the data from PLC, and then updates them in OPC server so that the TE process could get them.

2) *TE Process (in PC1)*: To enable the TE process in PC1 to communicate with OPC server and PLC, two toolboxes of Simulink are plugged into the process model: 1) OPC toolbox for OPC protocol, and 2) Modbus-Matlab-Simulator for Modbus protocol.

⁴<http://snap7.sourceforge.net/>



(a)



(b)

Fig. 4. The TE process (a) and controller (b)

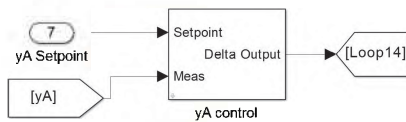


Fig. 5. The yA control-loop

These two toolboxes all provide the interfaces to read and write data from network. In TE process, we use these interfaces to replace the control-loop, an example of OPC toolbox is presented in Fig. 8. In contrast to Fig. 5, the OPC toolbox replaces the yA control-loop, and the inputs of yA control-loop are written to OPC toolbox and the output is read from OPC toolbox.

IV. THE ATTACKS SIMULATED IN VTET

The cyber attack to ICS is proved to be available by exploiting the vulnerabilities in communication protocols and PLC [6], [7]. Due to the fidelity to a real-world ICS, most

```
1 CASE #ssMethodType OF
2 @1
3
4 (* InitializeConditions for UnitDelay: '<S2>/Unit Delay' *)
5 #UnitDelay_DSTATE := 0.0;
6
7
8 (* Sum: '<S2>/Sum' incorporates:
9  * Import: '<Root>/yA'
10  * Import: '<Root>/yA Setpoint' *)
11 #rtb_Sum := #yASetpoint - #yA;
12
13 (* Output: '<Root>/Loop14' incorporates:
14  * Gain: '<S2>/Gain'
15  * Gain: '<S2>/Gain1'
16  * Sum: '<S2>/Sum1'
17  * UnitDelay: '<S2>/Unit Delay' *)
18 #Loop14 := ((0.1 * #rtb_Sum) + #rtb_Sum) - #UnitDelay_DSTATE * 0.0002;
19
20 (* Update for UnitDelay: '<S2>/Unit Delay' *)
21 #UnitDelay_DSTATE := #rtb_Sum;
22 END_CASE;
```

Fig. 6. The code of yA control-loop

Tag Name	Address	Data Type	Scan Rate	Scaling	Description
Loop14	MD10	DWord	100	None	
yASetpoint	MD2	DWord	100	None	
yA	MD5	DWord	100	None	

Fig. 7. The variables configuration in OPC sever

cyber attacks disclosed could be performed in VTET such as reconnaissance, DoS (Denial-of-Service) and the PLC program tampering.

Specially, in VTET we can simulate the attacks targeting the process to damage the physical devices which are impractical in a real-world environment. These attacks are launched by tampering the program in PLC or the data of actuators and sensors. The flow of tampering attack can refer to [8]. In the rest of the section, we will demonstrate the 5 attacks presented in VTET.

A. Attack 1: Exploding The Reactor

The reactor in TE should be cooled during the reaction process. The flow of cooling water is controlled using a PID control program. The attack decreases 1.1% of the integration coefficient to reduce the cooling water supply. As a result of the attack, the temperature and pressure of the reactor will increase and the reactor will explode at last. Fig. 9 shows that after the attack is launched (after 10 hours), the temperature and pressure of the reactor will increase but will descend quickly since the reactor has been exploded.

B. Attack 2: Reducing The Productivity

The attack bypasses the minimum productivity limitation of TE, and reduces the productivity by 35%. As shown in Fig.

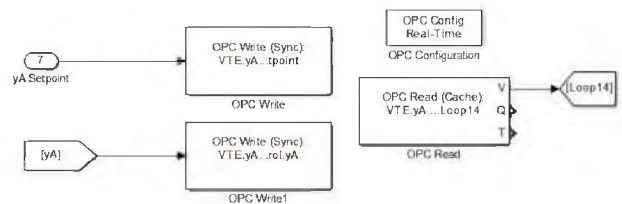


Fig. 8. Using OPC toolbox to read and write variables in TE

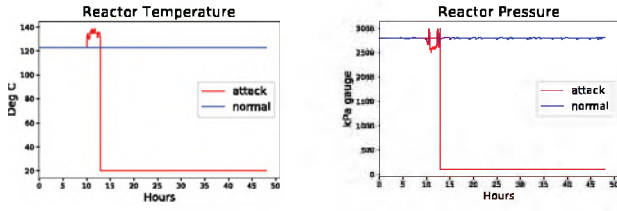


Fig. 9. The data of Attack 1

10, the feeds of reactants A, C, D and E decrease after the attack launched (after 15 hours). As the result of the attack, the product in the separator will be reduced.

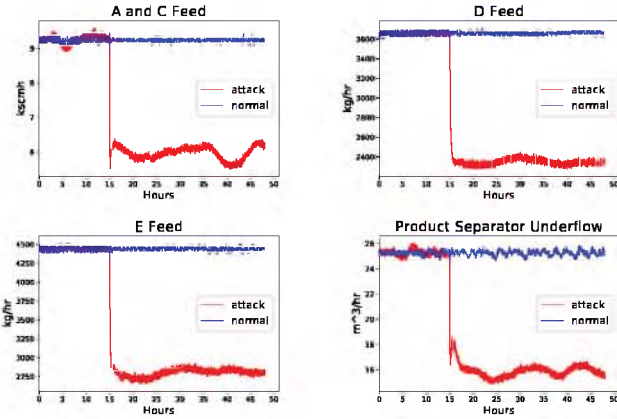


Fig. 10. The data of Attack 2

C. Attack 3: Jamming The Stripper

In the original TE process, the outlet valve of separator would open when the flow is lower than a setpoint, and close when the flow reaches the setpoint. The attack would cause the valve to close when the flow is lower than the setpoint. Fig. 11(a) shows that the stripper level will decrease to 0 when the attack occurs after 20 hours. In the meanwhile, the PLC will increase the feeds of reactants A, C, D and E since the flow of the product has decreased, the data is shown in Fig. 11(b), (c) and (d). This will cause the pressure of reactor and separator to increase, and the reactor and separator will explode finally, as shown in Fig. 11(e) and (f).

D. Attack 4: Increasing The Byproduct

The attack manipulates the inlet valves of A and C to decrease the feeds by 40%. According to the TE process equations in Section III-A, the quantities of product G and H are related to the feeds of A and C. Then the products will reduce and the byproduct F will increase. The decrease of stripper level will cause the system to increase the feeds of D and E as shown in Fig. 12(a), (b) and (c). The accumulation of the byproduct F will increase the pressure in both reactor and separator, and cause the explosion of them as shown in Fig. 12(d), (e) and (f).

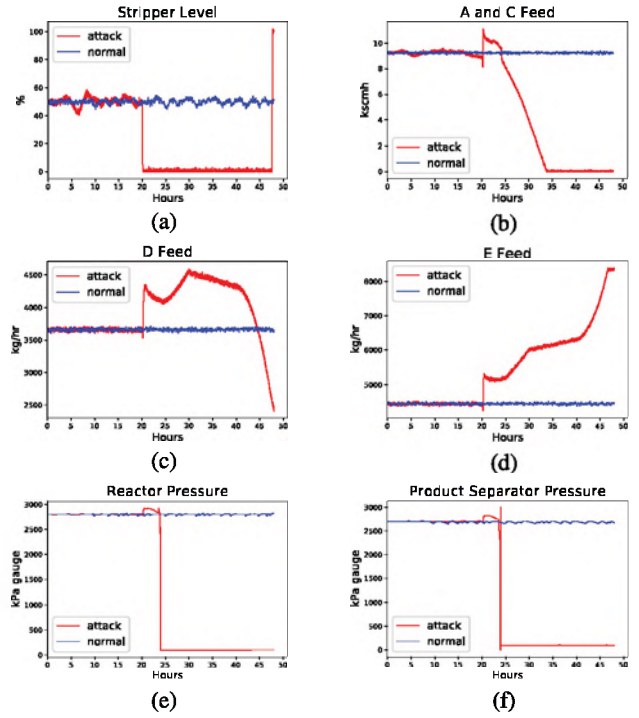


Fig. 11. The data of Attack 3

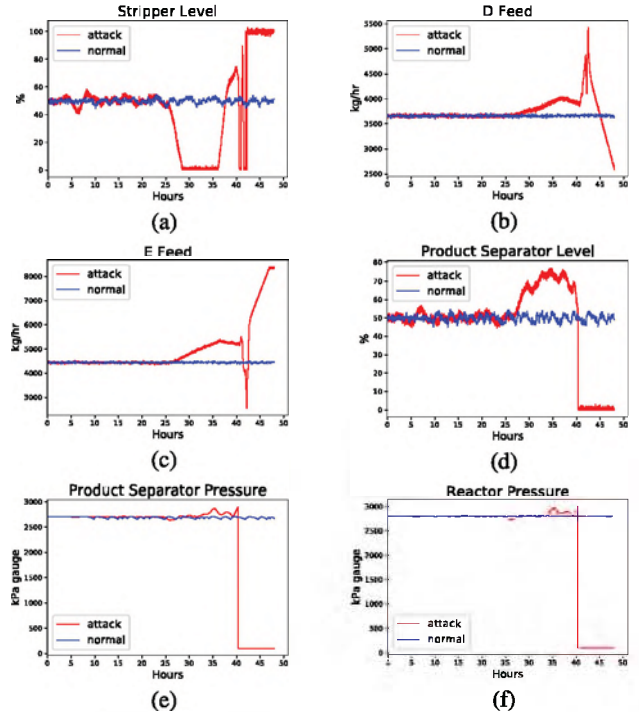


Fig. 12. The data of Attack 4

E. Attack 5: Disturbing The Process

The attack tempers the measurement of stripper temperature by decreasing 20 degree. It would cause the decrease of the cooling water supply. But the actual temperature of the separator and reactor will increase and then fluctuate as shown in Fig. 13(a) and (b). The pressure fluctuation also occurs in separator and reactor as shown in Fig. 13(c) and (d). Fig. 13(e) and (f) show that the level and underflow of separator (the products) will fluctuate even more dramatically. It will cause the system to be unstable and reduce the productivity of the process.

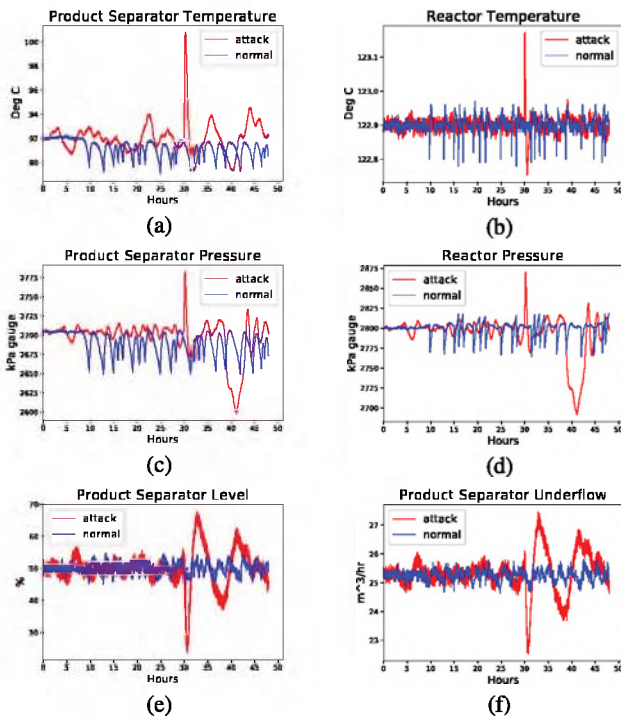


Fig. 13. The data of Attack 5

V. RELATED WORKS

In this section, we present an overview on the state-of-the-art of the ICS testbed.

Since the physical replicated ICS testbeds could achieve the best fidelity, they are widely used for researching and experimenting the cyber security technologies in a specific target. The National SCADA Test Bed (NSTB) [9] developed by U.S. Department of Energy is a large and full-scale electrical power grid which contains 61 miles transmission loop, distribution lines, and seven substations.

Although NSTB could provide a real-world testbed for multi-level cyber security research, it is still unfeasible to deploy this large testbed in the laboratorial environment. The testbeds scaling down a real-world system are introduced for academic usage. The SWaT [10] and WADI [11] designed by Singapore University of Technology and Design are two typical scale-down testbeds. The former is used for water

treatment process and the latter is used for water distribution. The development team of these testbeds also provides the datasets of the experiments for researchers⁵.

Some testbeds hybridize the physical and virtual (or emulated) components to make a trade-off between fidelity and economy.

The PowerCyber testbed [12] at Iowa State University integrates SCADA hardware and software along with emulation and simulation techniques to provide an accurate electric grid cyber infrastructure. Anirudh Pulella federates the PowerCyber and the DETER testbed [13] to provide wide-area realtime cyber security experimentation [14].

The EPIC testbed presented by Christos Siaterlis et al. [15] employs an Emulab-based technology to recreate the cyber part of system and multiple software simulators for the physical part. The testbed includes multiple process scenarios such as power plant, chemical production and electrical grid.

The NIST (National Institute of Standards and Technology, U.S.) develop a testbed which simulates the TE process [16]. This testbed is very similar to our work VTET. But the NIST testbed is a full-simulating system of which the controller is simulated by a PC, while VTET uses the physical PLC to achieve more fidelity.

VI. CONCLUSION

In this paper, we present a virtual ICS testbed VTET. The control target is a virtual chemical process (TE process) simulated by Matlab. The controller in VTET could be physical PLC or a simulator. 3 common ICS protocols (OPC, Modbus and Siemens S7) are support in VTET for network communication. We also demonstrate the implements of the testbed including the controller migration, system and network configuration. Specially, we simulate 5 attacks targeting the process to damage the physical devices to prove the availability of VTET. These attacks are impractical in a real-world ICS. The attack methods and the acquired data could be used for developing cyber security technologies for ICS. VTET is easy to configure and construct, so it is very suitable for preliminary ICS security research in laboratorial environment.

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